- **While Loop:-**

A while loop executes a block of code repeatedly as long as a condition is true.

Syntax → while (condition) {
            Statements
        }

NOTE:- (i) while loop is an <u>entry controlled loop</u>.
(ii) Condition is checked before execution.

- **Do While Loop:-**

A do-while loop executes the code block at least once, and then checks the condition.

Syntax → do {
            Statements
        } while (condition)

NOTE:- (i) Do-while loop is an <u>exit controlled loop</u>.
(ii) Condition is checked after execution.

- **Break Statement:-**

The break statement is used to exit a loop before it finishes all iterations.

Syntax → break ;

When condition for break becomes true, break executes and the loop ends.

• Continue Statement:-

The continue statement skips the current iteration and moves directly to the next iteration.

Syntax → continue;

• Switch Statement:-

A switch statement is used for multi-way decision making.

Syntax →
```
switch (expression) {
        case value1:
                code
                break;

        case value 2:
                code
                break;

        default:
                code
}
```

Working:-
(i) Expression is evaluated.
(ii) Its value is compared with each case.
(iii) If a matching case is found, that code executes
(iv) break terminates switch
(v) If no case matches, default executes.

• Scope of Variable:-

The scope of a variable determines where that variable can be accessed in a program.

> Local Scope
A variable declared inside a block {} can only be accessed within that block.

> Global Scope
A variable declared outside all functions can be accessed throughout the program.

> Variable Shadowing → A variable declared inside a block can have the same name as an outer variable.

Q) Input an integer n, find if it is prime or not using while loop and break.

```
#include <iostream>
using namespace std;
int main () {
    int n, i=2, flag=0;
    cin >> n
  while (i <n) {
        if (n%i ==0) {
            flag++;
            break;
        }
        i++;
    }
  if (flag==0)
        cout << "Prime" << endl;
  else
        cout << "Not Prime" << endl;
  return 0;
}
```

Q) Print all the Capital and Small letters using a while loop.

```
#include <iostream>
using namespace std;
int main () {
    char Series1 = 'A', Series2='a';
```

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```
cout << "\nCapital Letters: ";
while (Series1 <= 'Z') {
        cout << Series1 << " ";
        Series1++;
    }

cout << "\nSmall Letters: ";
while (Series2 <= 'z') {
        cout << Series2 << " ";
        Series2++;
    }
return 0;
}
```

Q) Given a number n, print all the numbers from 1 to n (inclusive) which are not divisible by 3 and 5. Using Continue.

```
#include <iostream>
using namespace std;

int main() {
    int n, i = 1;
    cin >> n;
  while (i <= n) {
        if (i % 3 == 0 || i % 5 == 0) {
            i++;   → i++ before continue is important.
            continue;
        }
        cout << i << " ";
        i++;
    }
  return 0;
}
```

Q) WAP to print the factorial of a number (n) using while and do-while loop.

```
#include <iostream>
using namespace std;
int main () {
        int n, i= 1, fact=1;
        cin >> n;
    while(i <=n) {
          fact *= i;
          i++;
     }
    cout << "Factorial (using while): " << fact << endl;
    fact= 1, i=1;
    do {
        fact *= i;
        i++;
     } while (i <=n)
    cout << " Factorial (using do-while): " << fact << endl;
    return 0;
 }
```

Q) WAP to print days of week using switch case.

```
#include <iostream>
using namespace std;
int main () {
        int n;
        cout << " Enter Day Number: " ;
        cin >> n;
```

```
switch (n) {
    case 1:  cout << "Monday" << endl;
             break;
    case 2:  cout << "Tuesday" << endl;
             break;
    case 3:  cout << "Wednessday" << endl;
             break;
    case 4:  cout << "Thursday" << endl;
             break;
    case 5:  cout << "Friday" << endl;
             break;
    case 6:  cout << "Saturday" << endl;
             break;
    case 7:  cout << "Sunday" << endl;
             break;
    default:  cout << "Invalid Input" << endl;
}
return 0;
}
```

- **Number System Conversion:-**

(A) **Decimal To Any Number System**

(i) Keep dividing the number by the base of required number System and Store the remainders.

(ii) The Converted number is obtained by reading remainders from bottom to top.

ex → Convert $(13)_{10}$ to Binary

2	13	
2	6	→ 1 Remainders
2	3	0
2	1	1
	0	1

So, $(13)_{10} = (1101)_2$

we keep dividing till the number becomes zero.